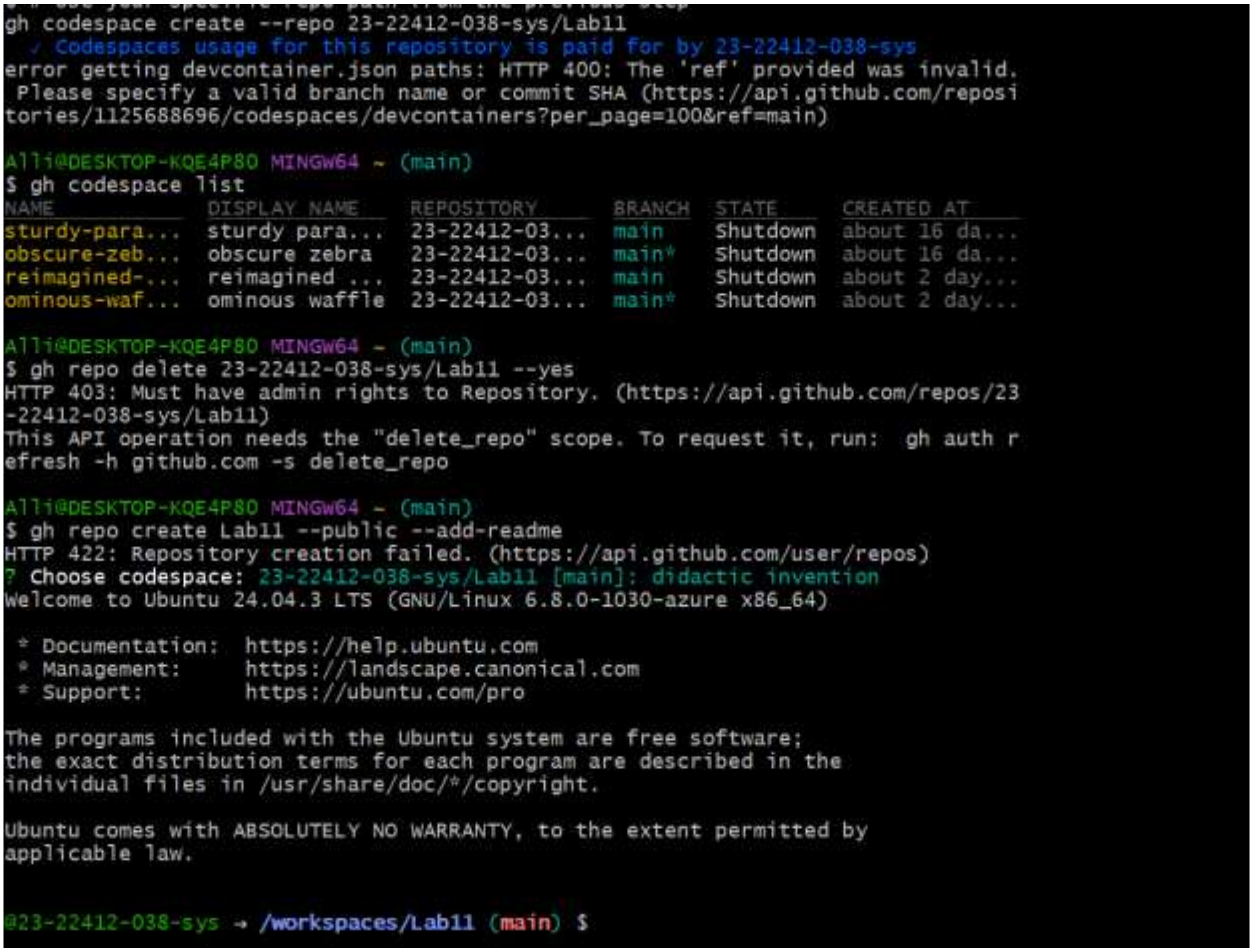
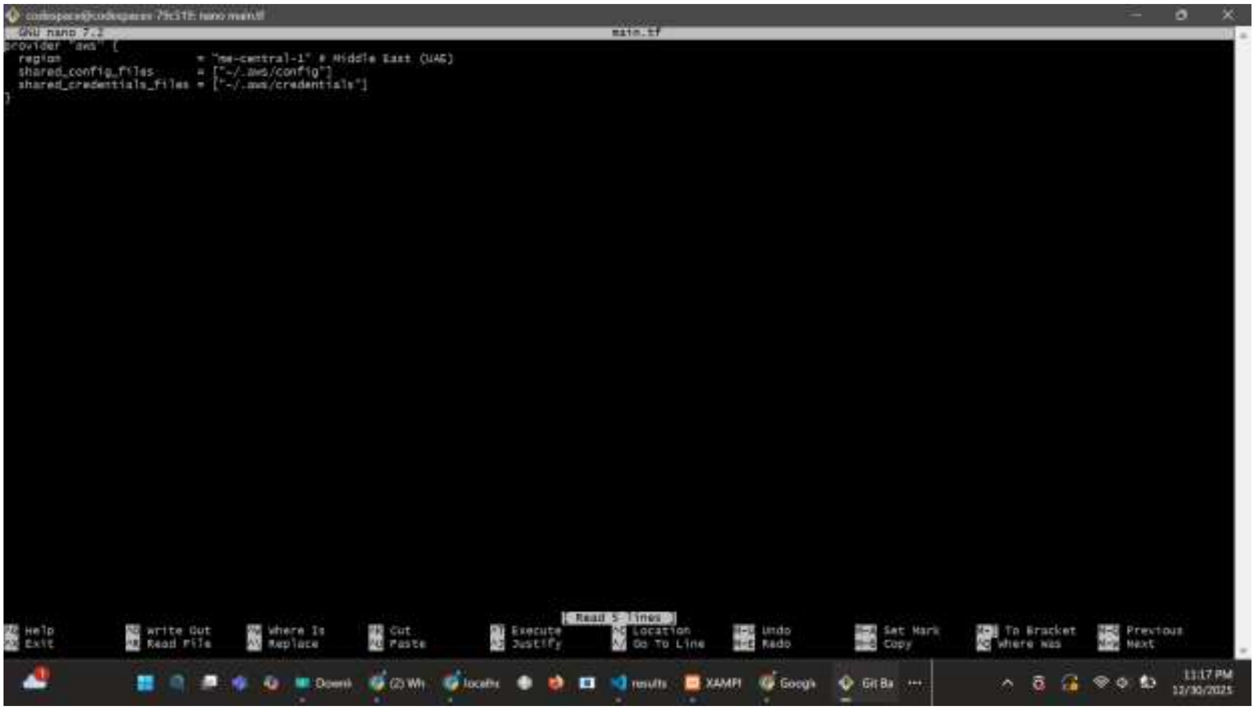
Task 0:

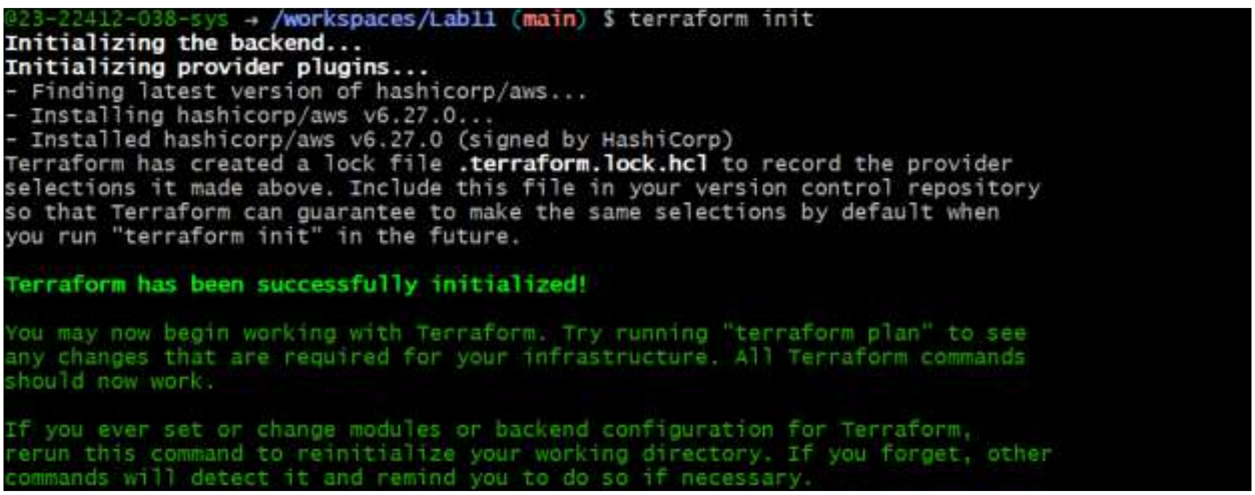


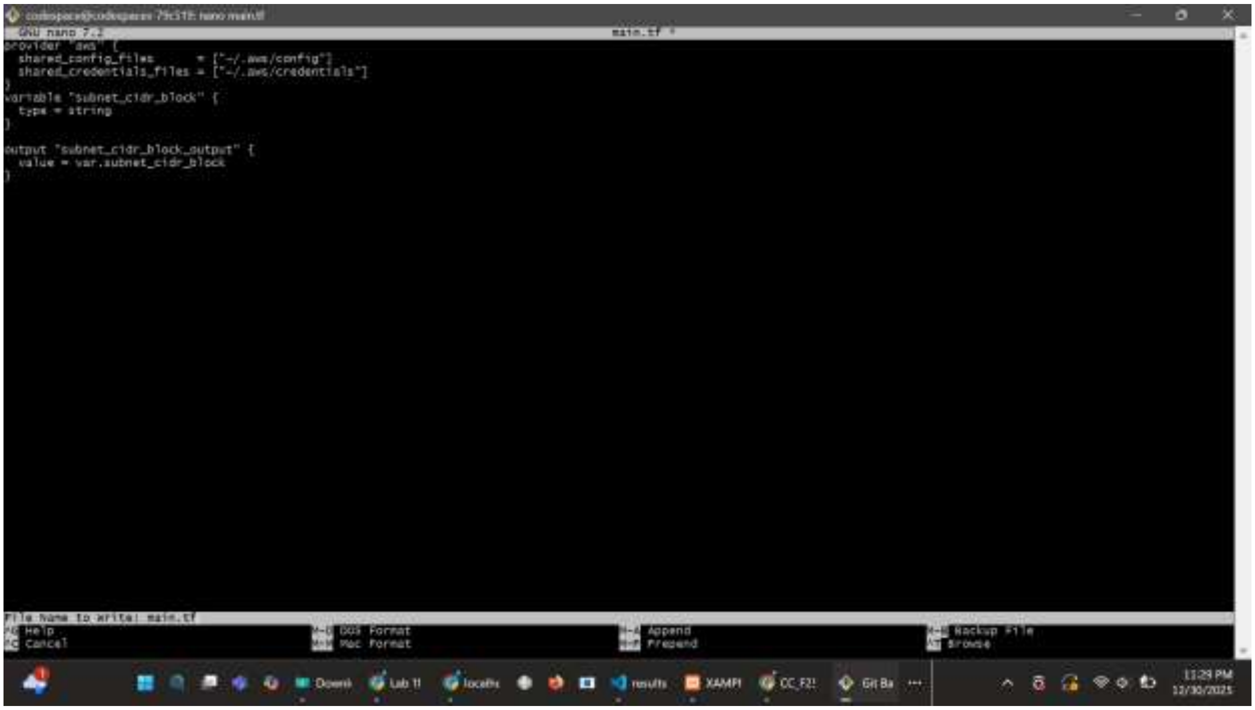
```
gh codespace create --repo 23-22412-038-sys/Lab11
  ✓ Codespaces usage for this repository is paid for by 23-22412-038-sys
error getting devcontainer.json paths: HTTP 400: The 'ref' provided was invalid.
 Please specify a valid branch name or commit SHA (https://api.github.com/reposi
tories/1125688696/codespaces/devcontainers?per_page=100&ref=main)

Alli@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ gh codespace list
NAME            DISPLAY NAME    REPOSITORY      BRANCH  STATE     CREATED AT
sturdy-para...  sturdy para...  23-22412-03...  main    Shutdown  about 16 da...
obscure-zeb...  obscure zebra   23-22412-03...  main*   Shutdown  about 16 da...
reimagined-...  reimagined ...  23-22412-03...  main    Shutdown  about 2 day...
ominous-waf...  ominous waffle  23-22412-03...  main*   Shutdown  about 2 day...

Alli@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ gh repo delete 23-22412-038-sys/Lab11 --yes
HTTP 403: Must have admin rights to Repository. (https://api.github.com/repos/23
-22412-038-sys/Lab11)
This API operation needs the "delete_repo" scope. To request it, run:  gh auth r
efresh -h github.com -s delete_repo

Alli@DESKTOP-KQE4P80 MINGW64 ~ (main)
$ gh repo create Lab11 --public --add-readme
HTTP 422: Repository creation failed. (https://api.github.com/user/repos)
? Choose codespace: 23-22412-038-sys/Lab11 [main]: didactic invention
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.


@23-22412-038-sys → /workspaces/Lab11 (main) $
```

Task 1

```
@23-22412-038-sys → /workspaces/Lab11 (main) $
@23-22412-038-sys → /workspaces/Lab11 (main) $ touch main.tf
@23-22412-038-sys → /workspaces/Lab11 (main) $ |
```


GNU nano 7.2
main.tf
provider "aws" {
region = "me-central-1" # Middle East (UAE)
shared_config_files = ["~/.aws/config"]
shared_credentials_files = ["~/.aws/credentials"]
}
[Read 5 lines]
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
Copy
To Bracket
Where Was
Previous
Next


@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.27.0...
- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
Terraform has been successfully initialized!
You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.
If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.


GNU nano 7.2
main.tf *
provider "aws" {
shared_config_files   = ["~/.aws/config"]
shared_credentials_files = ["~/.aws/credentials"]
}
variable "subnet_cidr_block" {
type = string
}
output "subnet_cidr_block_output" {
value = var.subnet_cidr_block
}
File Name to Write: main.tf
Help
Cancel
DOS Format
Mac Format
Append
Prepend
Backup File
Browse
11:29 PM
12/30/2025


/workspaces/Lab11 (main) $ terraform apply -auto-approve
var.subnet_cidr_block
Enter a value: 1
Changes to Outputs:
+ subnet_cidr_block_output = "1"
You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.
Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
Outputs:
subnet_cidr_block_output = "1"

```
GNU nano 7.2                                   main.tf *
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
variable "subnet_cidr_block" {
  type = string
default = "10.0.0.0/24"

}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}
```

```
^G Help      ^O Write Out  ^W Where Is  ^K Cut      ^T Execute  ^C Location    M-U Undo  M-A Set Mark   M-] To Bracket  M-Q Previous
^X Exit      ^R Read File  ^\ Replace   ^U Paste    ^J Justify  ^/ Go To Line  M-E Redo  M-6 Copy       ^Q Where Was    M-W Next
```

```
923-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  ~ subnet_cidr_block_output = "1" -> "10.0.0.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

subnet_cidr_block_output = "10.0.0.0/24"
```

```
923-22412-038-sys → /workspaces/Lab11 (main) $ export TF_VAR_subnet_cidr_block=10.0.20.0/24
terraform apply -auto-approve

Changes to Outputs:
  ~ subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.20.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

subnet_cidr_block_output = "10.0.20.0/24"
```

```
923-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  ~ subnet_cidr_block_output = "10.0.20.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

subnet_cidr_block_output = "10.0.30.0/24"
```

```
23-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0.40.0/24"

hanges to Outputs:
 ~ subnet_cidr_block_output = "10.0.30.0/24" -> "10.0.40.0/24"

ou can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

pply complete! Resources: 0 added, 0 changed, 0 destroyed.

utputs:

ubnet_cidr_block_output = "10.0.40.0/24"
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ printenv | grep TF_VAR_
unset TF_VAR_subnet_cidr_block
printenv | grep TF_VAR_
TF_VAR_subnet_cidr_block=10.0.20.0/24
```

Task 2

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0.0.0/24"

Changes to Outputs:
  ~ subnet_cidr_block_output = "10.0.40.0/24" -> "10.0.0.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

subnet_cidr_block_output = "10.0.0.0/24"
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "api_session_token=my_API_session_Token_12345"

Changes to Outputs:
  + api_session_token_output = (sensitive value)
  ~ subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

api_session_token_output = <sensitive>
subnet_cidr_block_output = "10.0.30.0/24"
```

```
GNU nano 7.2                    terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 7,
  "lineage": "0d7ac30c-597d-354f-956a-9b5b80f74f35",
  "outputs": {
    "api_session_token_output": {
      "value": "my_API_session_Token_12345",
      "type": "string",
      "sensitive": true
    },
    "subnet_cidr_block_output": {
      "value": "10.0.30.0/24",
      "type": "string"
    }
  },
  "resources": [],
  "check_results": [
    {
      "object_kind": "var",
      "config_addr": "var.api_session_token",
      "status": "pass",
      "objects": [
        {
          "object_addr": "var.api_session_token",
          "status": "pass"
        }
      ]
    },
    {
      "object_kind": "var",
      "config_addr": "var.subnet_cidr_block",
      "status": "pass",
      "objects": [
        {
          "object_addr": "var.subnet_cidr_block",
          "status": "pass"
        }
      ]
    }
  ]
}
                          [ Wrote 42 lines ]
^G Help      ^O Write Out  ^W Where Is  ^K Cut    ^T Execute  ^C Location    M-U Undo  M-A Set Mark  M-] To Bracket  M-Q Previous
^X Exit      ^R Read File  ^\ Replace   ^U Paste  ^J Justify  ^/ Go To Line  M-E Redo  M-6 Copy      ^Q Where Was    M-W Next
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "api_session_token=my_API_session_Token_123456"

Changes to Outputs:
  + api_session_token_output = (sensitive value)
  ~ subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.
```

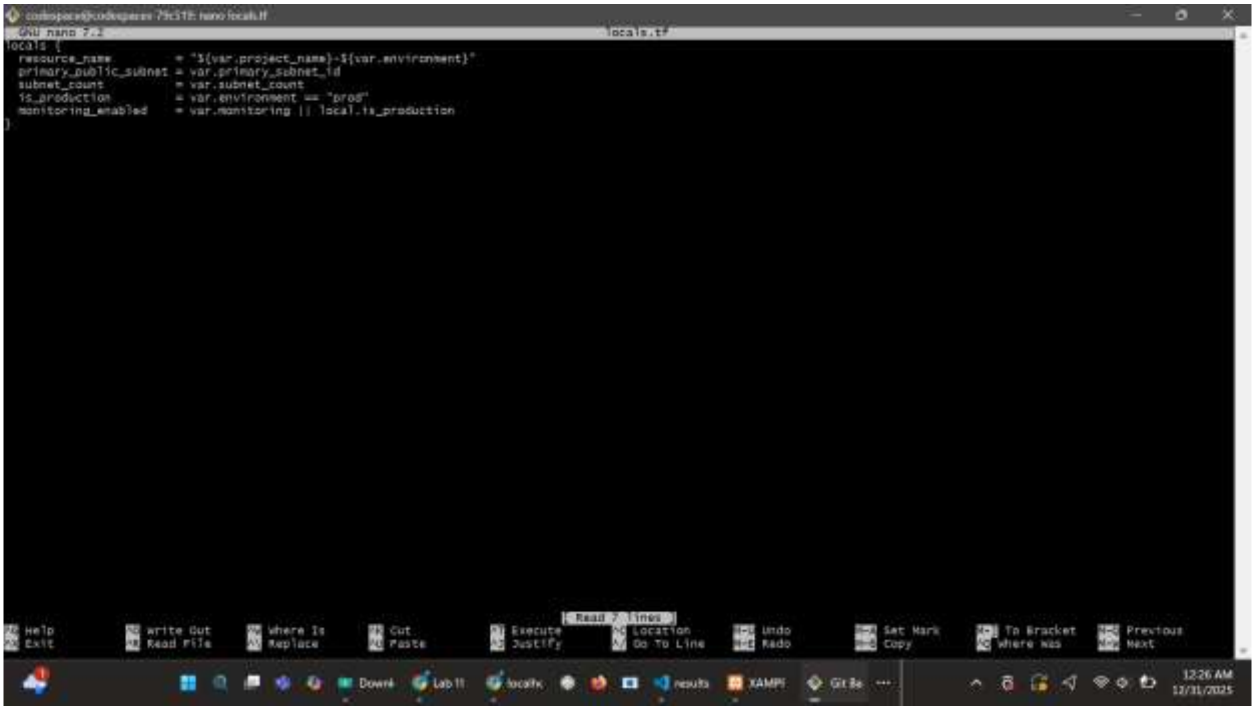
```
@23-22412-038-sys → /workspaces/Lab11 (main) $ cat terraform.tfstate | grep api_session_token -n
7:    "api_session_token_output": {
21:      "config_addr": "var.api_session_token",
25:          "object_addr": "var.api_session_token",
```

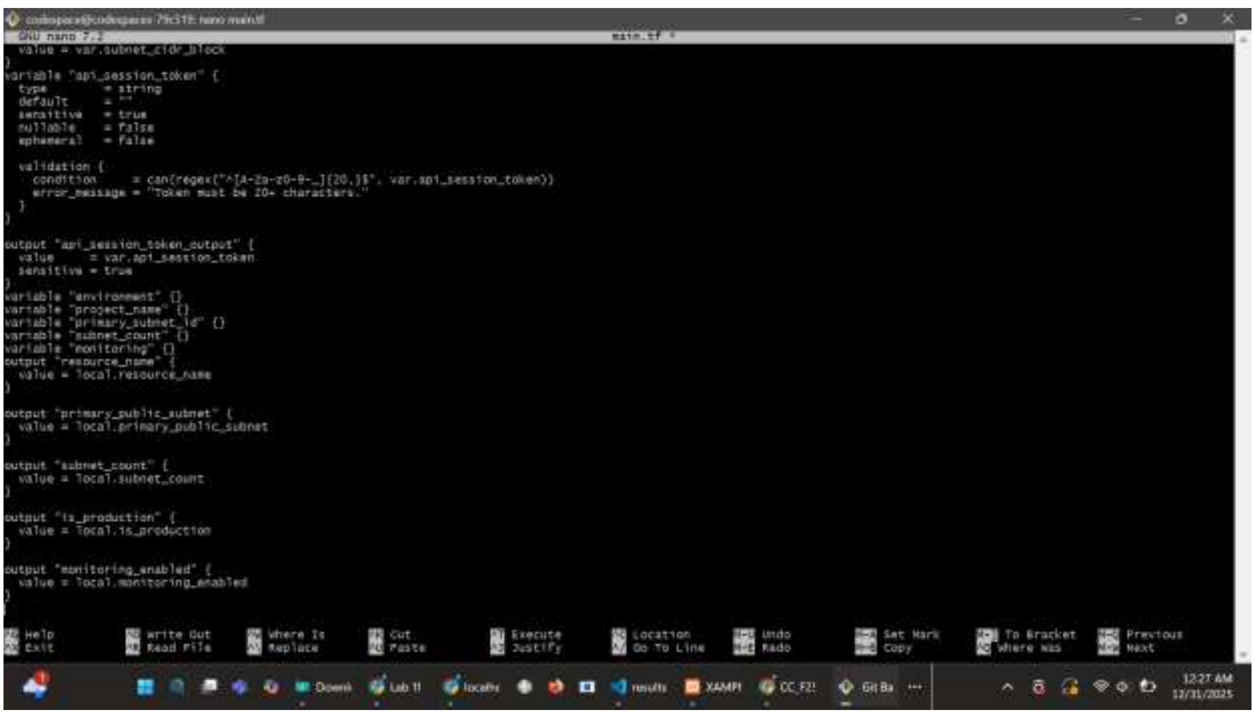
Task 3

```
GNU nano 7.2                                   main.tf
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
variable "subnet_cidr_block" {
  type        = string
  default     = ""
  description = "CIDR block to assign to the application subnet"
  sensitive   = false
  nullable    = false
  ephemeral   = false

  validation {
    condition     = can(regex("^([0-9]{1,3}\\.){3}[0-9]{1,3}/[0-9]+$", var.subnet_cidr_block))
    error_message = "The subnet_cidr_block must be a valid CIDR notation string."
  }
}


output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}
variable "api_session_token" {
  type        = string
  default     = ""
  sensitive   = true
  nullable    = false
  ephemeral   = false

  validation {
    condition     = can(regex("^[A-Za-z0-9-_]{20,}$", var.api_session_token))
    error_message = "Token must be 20+ characters."
  }
}

output "api_session_token_output" {
  value     = var.api_session_token
  sensitive = true
}
variable "environment" {}
variable "project_name" {}
variable "primary_subnet_id" {}
variable "subnet_count" {}
variable "monitoring" {}
                                   [ Read 47 lines ]
^G Help      ^O Write Out  ^W Where Is  ^K Cut      ^T Execute  ^C Location   M-U Undo  M-A Set Mark  M-] To Bracket  M-Q Previous
^X Exit      ^R Read File  ^\ Replace   ^U Paste    ^J Justify  ^/ Go To Line M-E Redo  M-6 Copy      ^Q Where Was     M-W Next
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ aws ec2 describe-subnets \
  --filters "Name=availability-zone,Values=me-central-1a" \
  --query "Subnets[].SubnetId" \
  --output text
subnet-08a73f8aa1dee72d3
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ cat terraform.tfvars
subnet_cidr_block = "10.0.30.0/24"
environment         = "dev"
project_name        = "lab_work"
primary_subnet_id   = "subnet-08a73f8aa1dee72d3
"
subnet_count        = 3
monitoring          = true
@23-22412-038-sys → /workspaces/Lab11 (main) $ |
```


GNU nano 7.2
locals.tf
locals {
resource_name = "${var.project_name}-${var.environment}"
primary_public_subnet = var.primary_subnet_id
subnet_count = var.subnet_count
is_production = var.environment == "prod"
monitoring_enabled = var.monitoring || local.is_production
}
[Read 7 lines]
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
Copy
To Bracket
Where Was
Previous
Next
12:26 AM
12/31/2025


GNU nano 7.2
main.tf *
value = var.subnet_cidr_block
}
variable "api_session_token" {
type = string
default = ""
sensitive = true
nullable = false
ephemeral = false
validation {
condition = can(regex("^[A-Za-z0-9-_]{20,}$", var.api_session_token))
error_message = "Token must be 20+ characters."
}
}
output "api_session_token_output" {
value = var.api_session_token
sensitive = true
}
variable "environment" {}
variable "project_name" {}
variable "primary_subnet_id" {}
variable "subnet_count" {}
variable "monitoring" {}
output "resource_name" {
value = local.resource_name
}
output "primary_public_subnet" {
value = local.primary_public_subnet
}
output "subnet_count" {
value = local.subnet_count
}
output "is_production" {
value = local.is_production
}
output "monitoring_enabled" {
value = local.monitoring_enabled
}
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
Copy
To Bracket
Where Was
Previous
Next
12:27 AM
12/31/2025

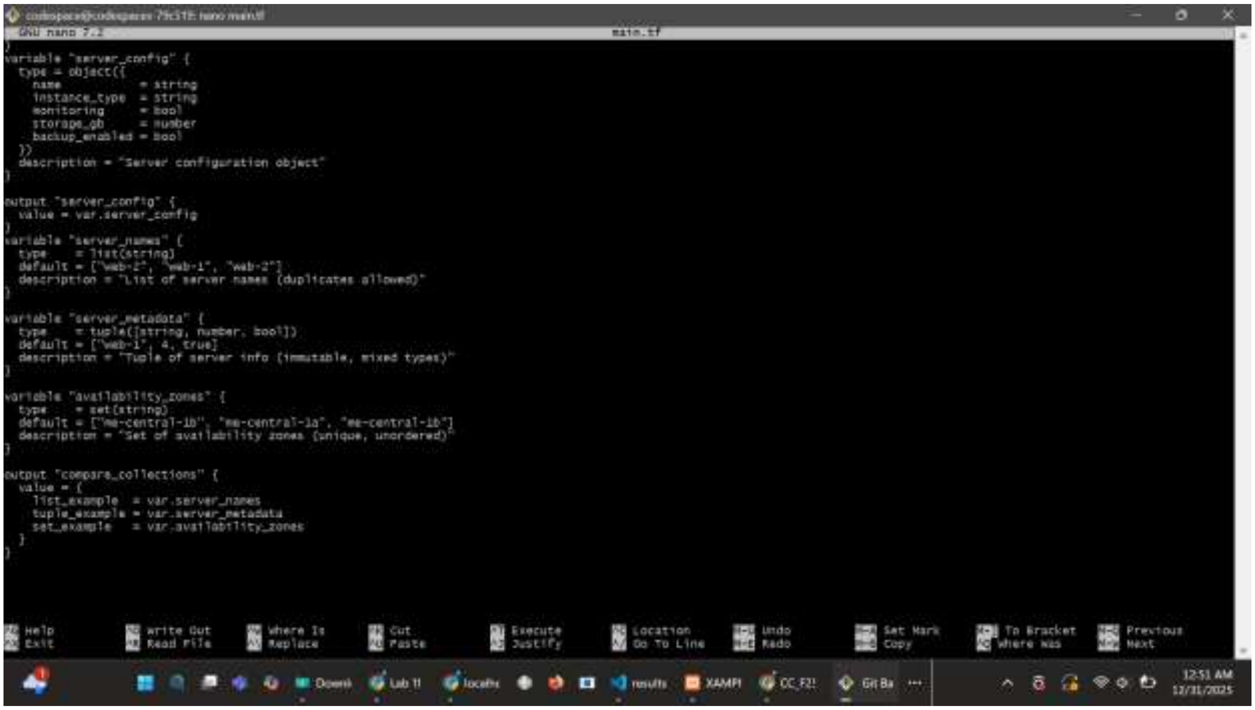
```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  ~ api_session_token_output = (sensitive value)
  + is_production            = false
  + monitoring_enabled       = true
  + primary_public_subnet    = "subnet-08a73f8aa1dee72d3"
  + resource_name            = "lab_work-dev"
  ~ subnet_cidr_block_output = "10.0.30.0/24" -> "10.0.0.0/24"
  + subnet_count             = 3

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

api_session_token_output = <sensitive>
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
```

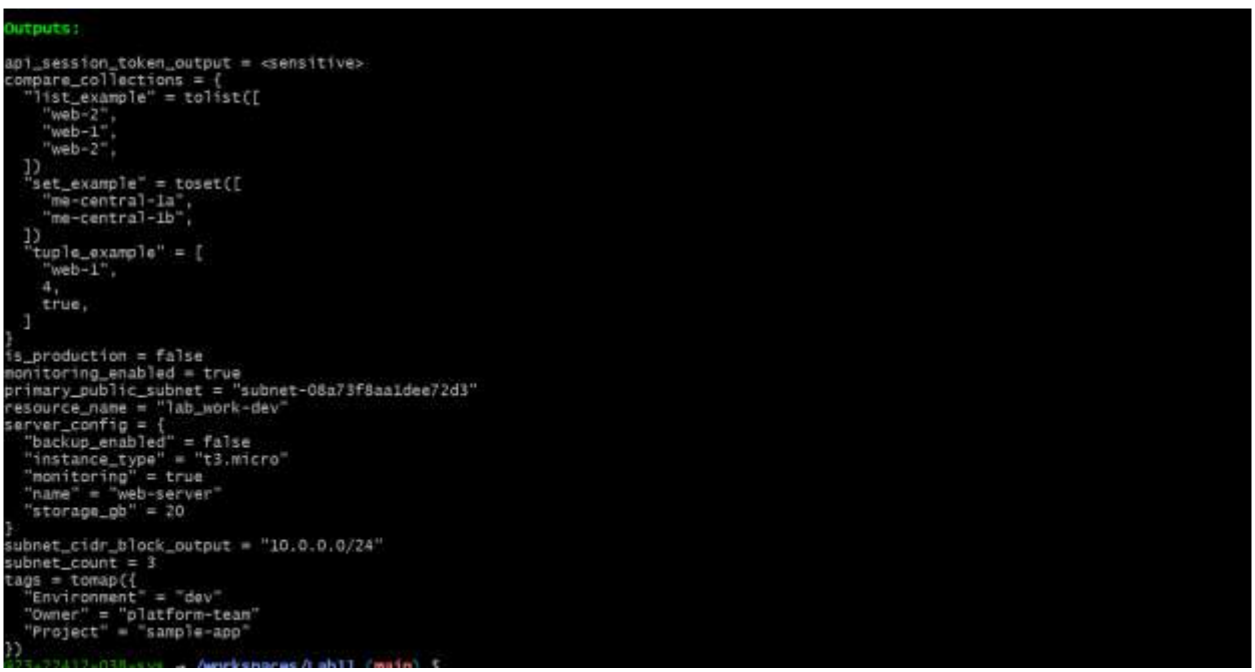
Task 4

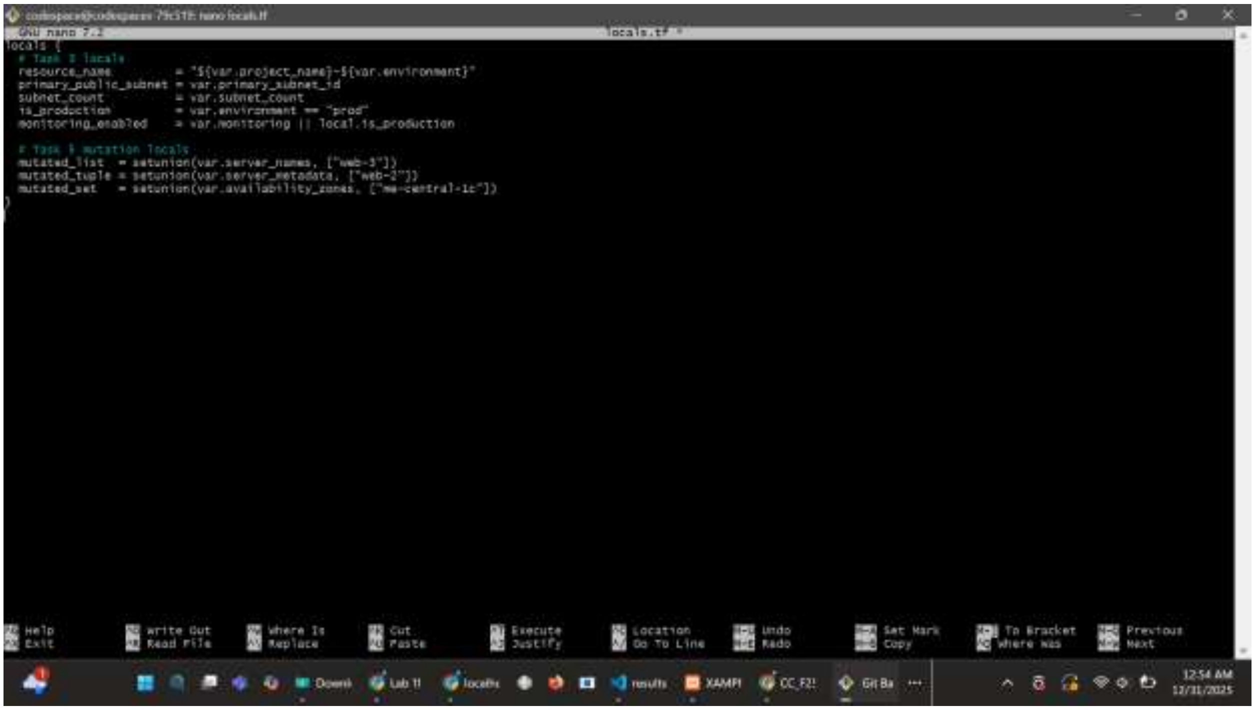
```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  + tags                     = {
      + Environment = "dev"
      + Owner       = "platform-team"
      + Project     = "sample-app"
    }

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

api_session_token_output = <sensitive>
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
})
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  + server_config            = {
      + backup_enabled = false
      + instance_type  = "t3.micro"
      + monitoring     = true
      + name           = "web-server"
      + storage_gb     = 20
    }

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

api_session_token_output = <sensitive>
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
})
@23-22412-038-sys → /workspaces/Lab11 (main) $
```

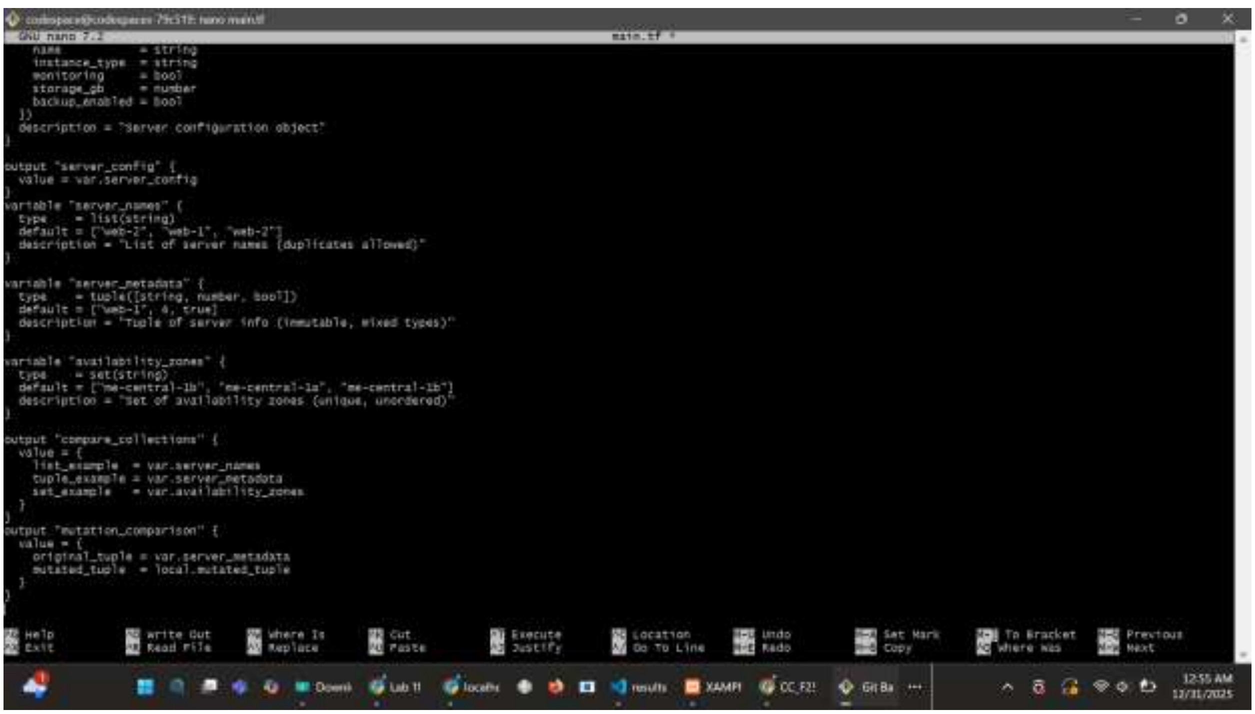
Task 5


GNU nano 7.2
main.tf
variable "server_config" {
type = object({
name = string
instance_type = string
monitoring = bool
storage_gb = number
backup_enabled = bool
})
description = "Server configuration object"
}
output "server_config" {
value = var.server_config
}
variable "server_names" {
type = list(string)
default = ["web-2", "web-1", "web-2"]
description = "List of server names (duplicates allowed)"
}
variable "server_metadata" {
type = tuple([string, number, bool])
default = ["web-1", 4, true]
description = "Tuple of server info (immutable, mixed types)"
}
variable "availability_zones" {
type = set(string)
default = ["me-central-1b", "me-central-1a", "me-central-1b"]
description = "Set of availability zones (unique, unordered)"
}
output "compare_collections" {
value = {
list_example = var.server_names
tuple_example = var.server_metadata
set_example = var.availability_zones
}
}
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
Copy
To Bracket
Where Was
Previous
Next
12:51 AM
12/31/2025


Outputs:
api_session_token_output = <sensitive>
compare_collections = {
"list_example" = tolist([
"web-2",
"web-1",
"web-2",
])
"set_example" = toset([
"me-central-1a",
"me-central-1b",
])
"tuple_example" = [
"web-1",
4,
true,
]
}
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
"backup_enabled" = false
"instance_type" = "t3.micro"
"monitoring" = true
"name" = "web-server"
"storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
"Environment" = "dev"
"Owner" = "platform-team"
"Project" = "sample-app"
})



```
locals {
  # Task 3 locals
  resource_name          = "${var.project_name}-${var.environment}"
  primary_public_subnet = var.primary_subnet_id
  subnet_count           = var.subnet_count
  is_production          = var.environment == "prod"
  monitoring_enabled     = var.monitoring || local.is_production

  # Task 5 mutation locals
  mutated_list  = setunion(var.server_names, ["web-3"])
  mutated_tuple = setunion(var.server_metadata, ["web-2"])
  mutated_set   = setunion(var.availability_zones, ["me-central-1c"])
}
```



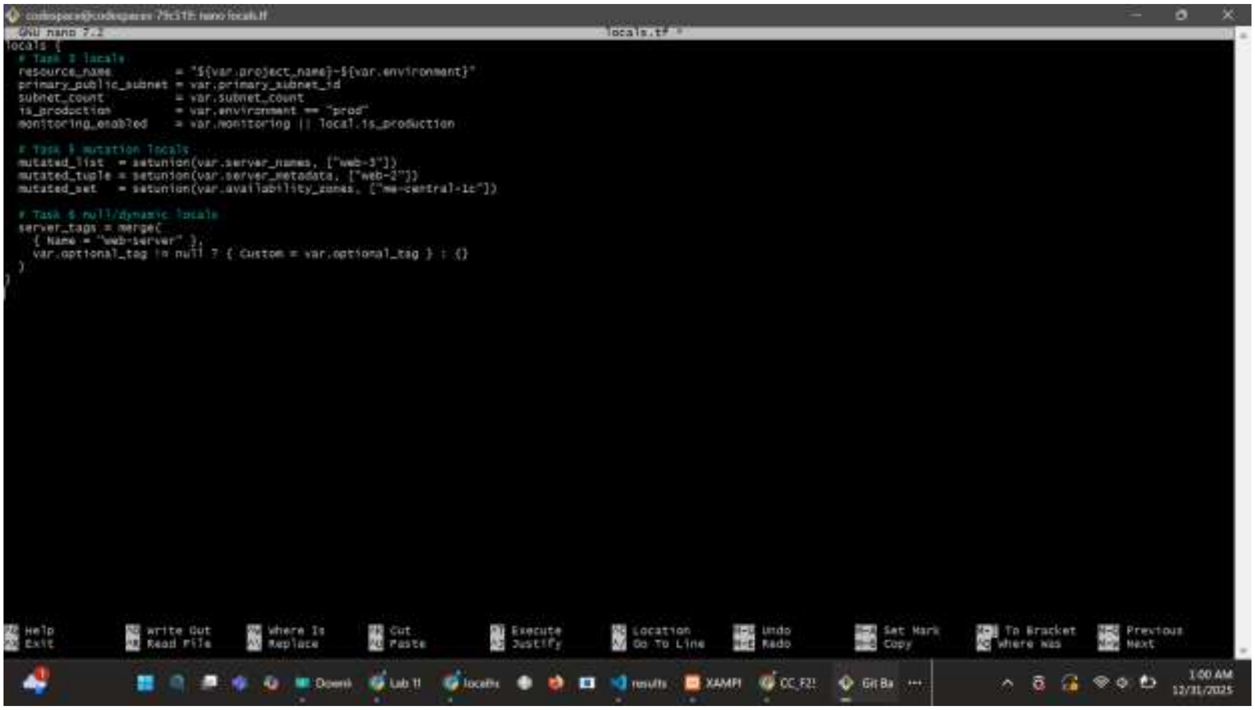
```
    name           = string
    instance_type  = string
    monitoring     = bool
    storage_gb     = number
    backup_enabled = bool
  })
  description = "Server configuration object"
}

output "server_config" {
  value = var.server_config
}
variable "server_names" {
  type    = list(string)
  default = ["web-2", "web-1", "web-2"]
  description = "List of server names (duplicates allowed)"
}

variable "server_metadata" {
  type    = tuple([string, number, bool])
  default = ["web-1", 4, true]
  description = "Tuple of server info (immutable, mixed types)"
}

variable "availability_zones" {
  type    = set(string)
  default = ["me-central-1b", "me-central-1a", "me-central-1b"]
  description = "Set of availability zones (unique, unordered)"
}

output "compare_collections" {
  value = {
    list_example  = var.server_names
    tuple_example = var.server_metadata
    set_example   = var.availability_zones
  }
}
output "mutation_comparison" {
  value = {
    original_tuple = var.server_metadata
    mutated_tuple  = local.mutated_tuple
  }
}
```

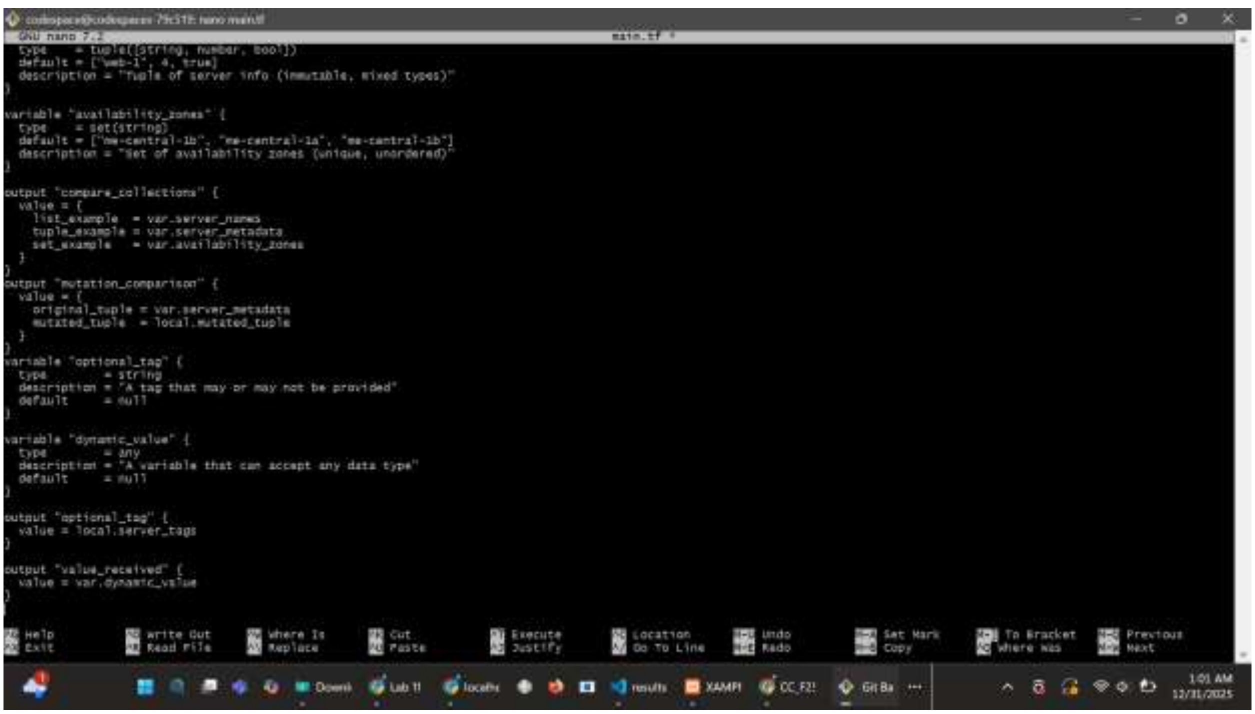
```
023-22412-036-sys → /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  + mutation_comparison     = {
      + mutated_tuple  = [
          + "4",
          + "true",
          + "web-1",
          + "web-2",
        ]
      + original_tuple = [
          + "web-1",
          + 4,
          + true,
        ]
    }

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.
```

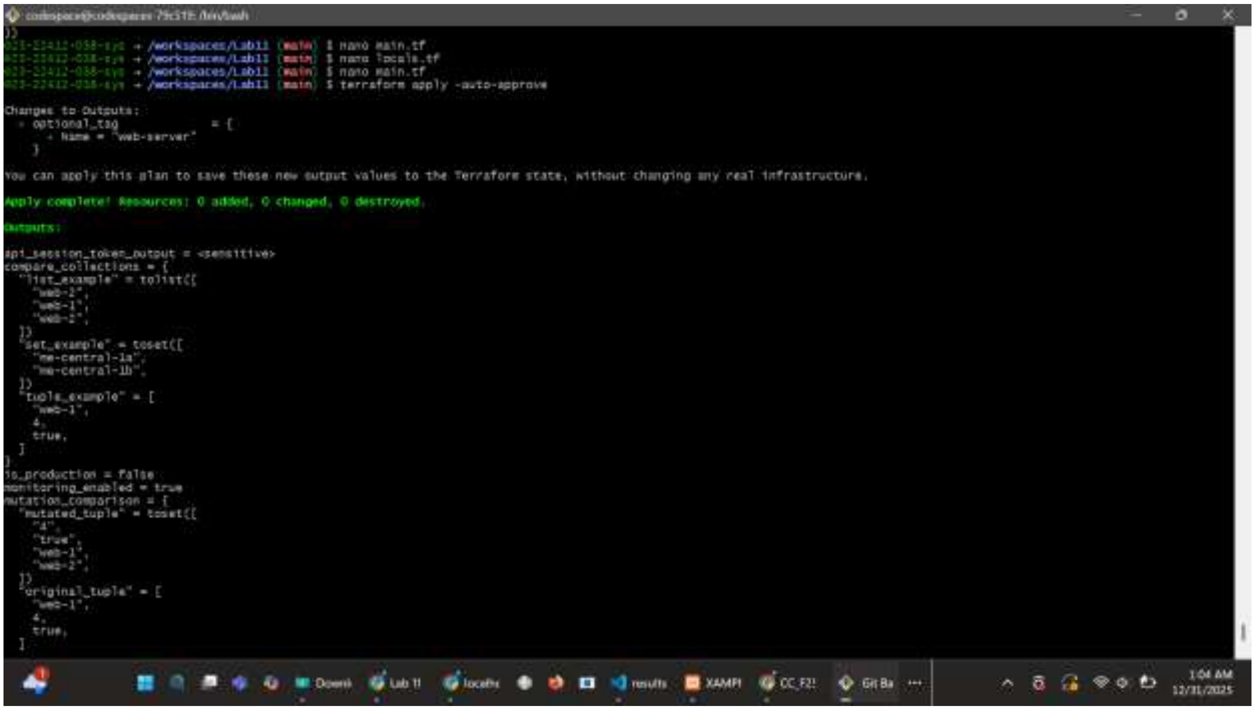
```
Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
```

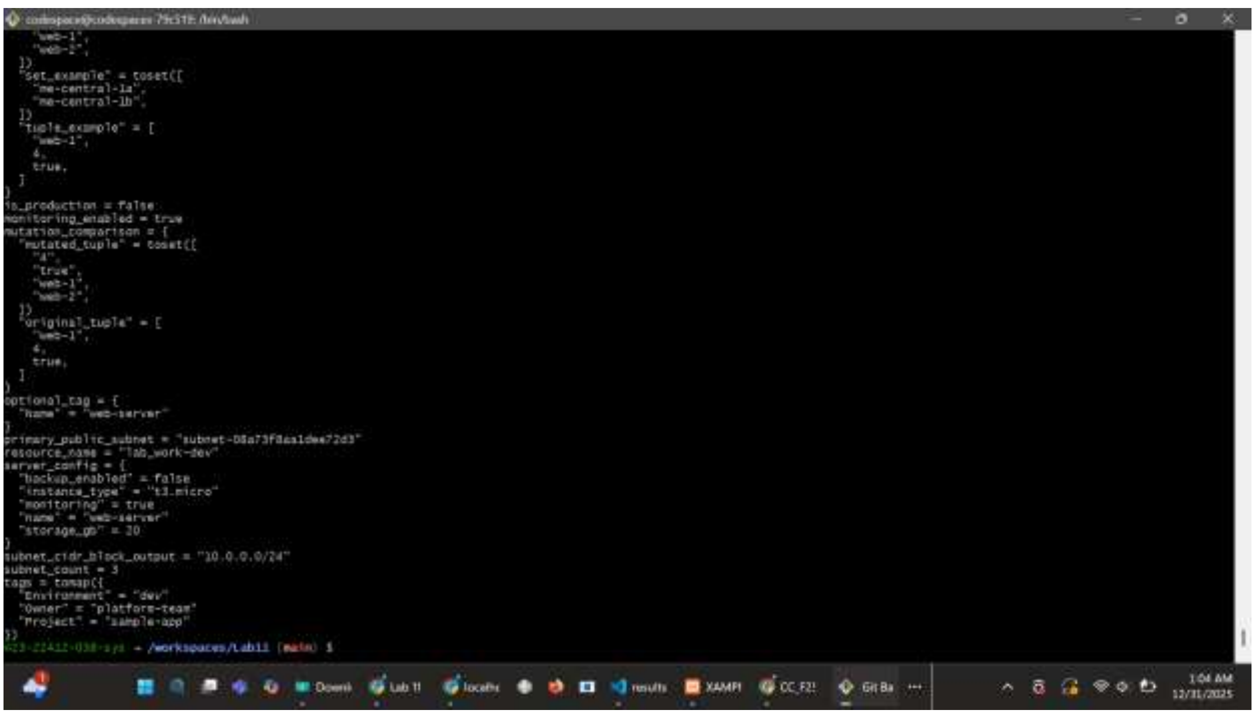
```
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
})
```

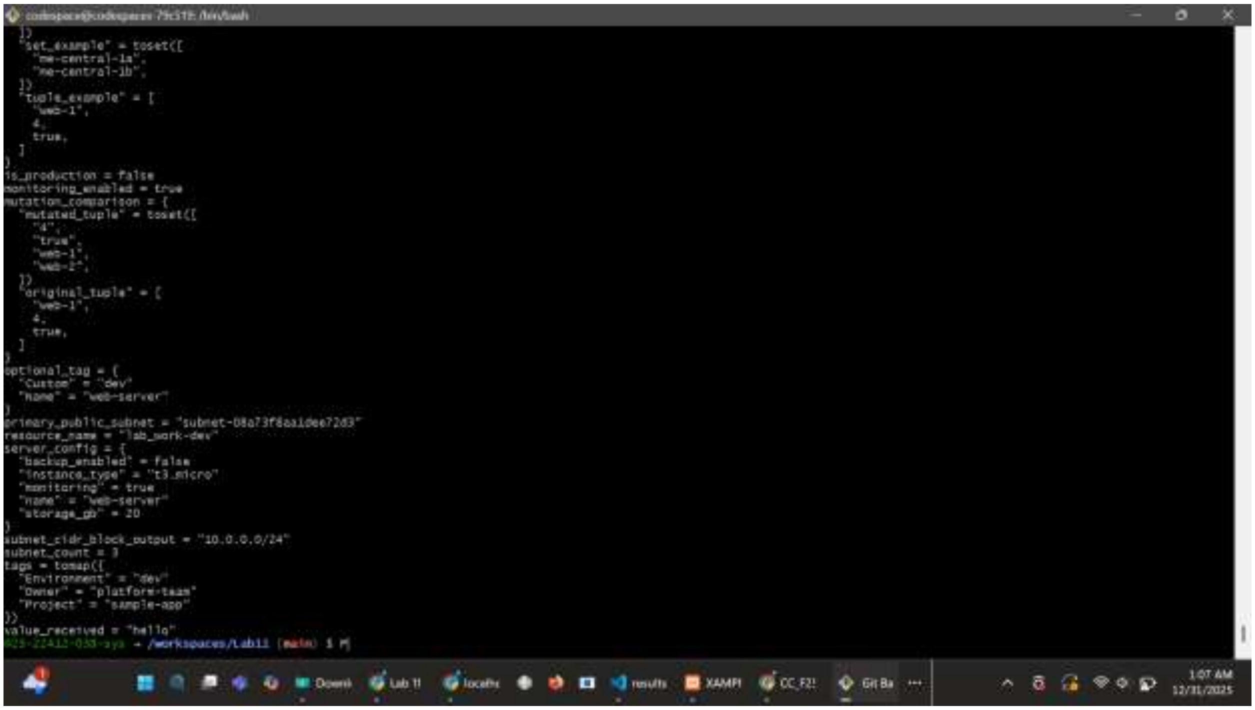
Task 6

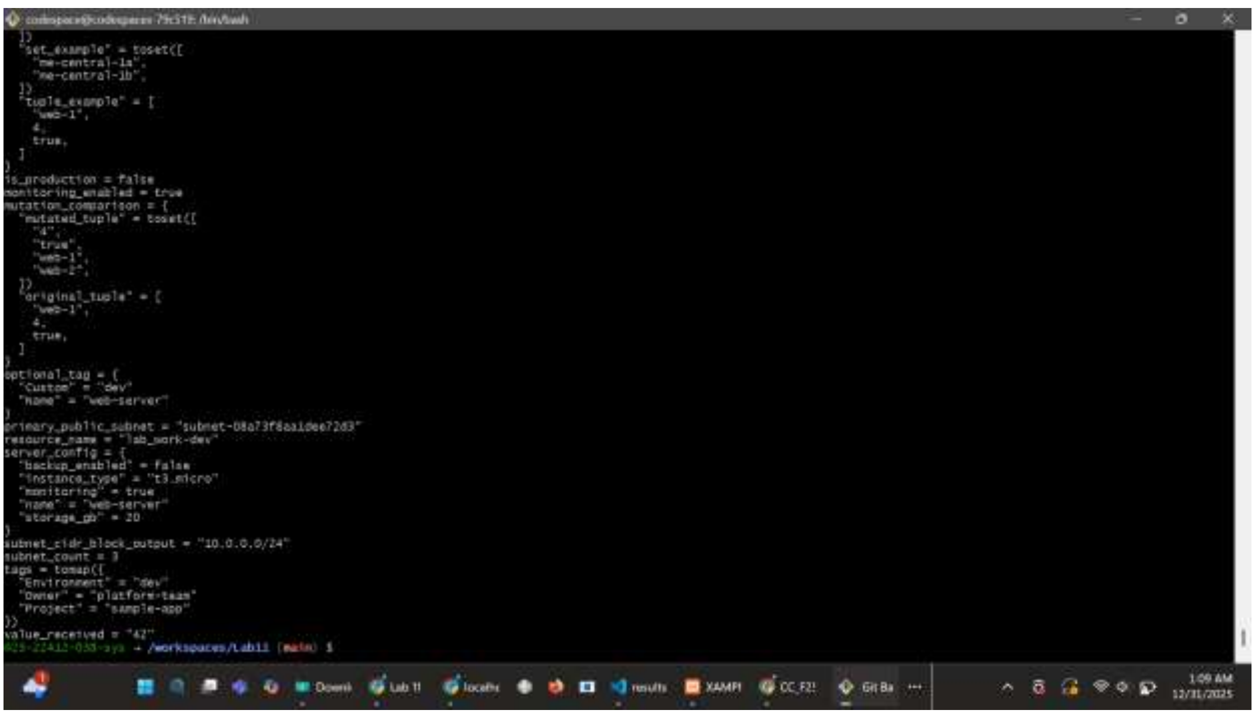

GNU nano 7.2
locals.tf *
locals {
# Task 3 locals
resource_name = "${var.project_name}-${var.environment}"
primary_public_subnet = var.primary_subnet_id
subnet_count = var.subnet_count
is_production = var.environment == "prod"
monitoring_enabled = var.monitoring || local.is_production
# Task 5 mutation locals
mutated_list = setunion(var.server_names, ["web-3"])
mutated_tuple = setunion(var.server_metadata, ["web-2"])
mutated_set = setunion(var.availability_zones, ["me-central-1c"])
# Task 6 null/dynamic locals
server_tags = merge(
{ Name = "web-server" },
var.optional_tag != null ? { Custom = var.optional_tag } : {}
)
}
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
Copy
To Bracket
Where Was
Previous
Next
1:00 AM
12/31/2025


GNU nano 7.2
main.tf *
type = tuple([string, number, bool])
default = ["web-1", 4, true]
description = "Tuple of server info (immutable, mixed types)"
}
variable "availability_zones" {
type = set(string)
default = ["me-central-1b", "me-central-1a", "me-central-1b"]
description = "Set of availability zones (unique, unordered)"
}
output "compare_collections" {
value = {
list_example = var.server_names
tuple_example = var.server_metadata
set_example = var.availability_zones
}
}
output "mutation_comparison" {
value = {
original_tuple = var.server_metadata
mutated_tuple = local.mutated_tuple
}
}
variable "optional_tag" {
type = string
description = "A tag that may or may not be provided"
default = null
}
variable "dynamic_value" {
type = any
description = "A variable that can accept any data type"
default = null
}
output "optional_tag" {
value = local.server_tags
}
output "value_received" {
value = var.dynamic_value
}
Help
Exit
Write Out
Read File
Where Is
Replace
Cut
Paste
Execute
Justify
Location
Go To Line
Undo
Redo
Set Mark
Copy
To Bracket
Where Was
Previous
Next
1:01 AM
12/31/2025


codespace@codespaces-79c51f: /bin/bash
$ nano main.tf
$ nano locals.tf
$ nano main.tf
$ terraform apply -auto-approve
Changes to Outputs:
optional_tag = {
+ Name = "web-server"
}
You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.
Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
Outputs:
api_session_token_output = <sensitive>
compare_collections = {
"list_example" = tolist([
"web-2",
"web-1",
"web-2",
])
"set_example" = toset([
"me-central-1a",
"me-central-1b",
])
"tuple_example" = [
"web-1",
4,
true,
]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
"mutated_tuple" = toset([
"4",
"true",
"web-1",
"web-2",
])
"original_tuple" = [
"web-1",
4,
true,
]
1:04 AM
12/31/2025

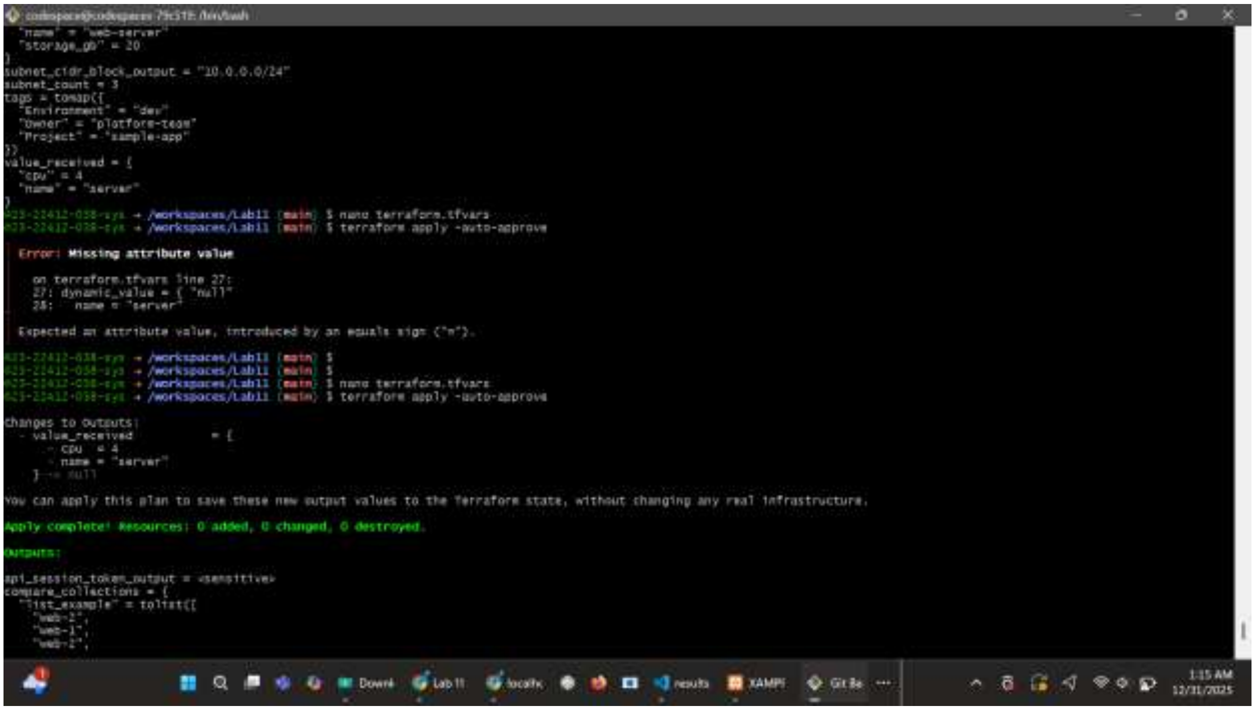

codespace@codespaces-79c51f: /bin/bash
"web-1",
"web-2",
])
"set_example" = toset([
"me-central-1a",
"me-central-1b",
])
"tuple_example" = [
"web-1",
4,
true,
]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
"mutated_tuple" = toset([
"4",
"true",
"web-1",
"web-2",
])
"original_tuple" = [
"web-1",
4,
true,
]
}
optional_tag = {
"Name" = "web-server"
}
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
"backup_enabled" = false
"instance_type" = "t3.micro"
"monitoring" = true
"name" = "web-server"
"storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
"Environment" = "dev"
"Owner" = "platform-team"
"Project" = "sample-app"
})
/workspaces/Lab11 (main) $
1:04 AM
12/31/2025


codespace@codespaces-79c51f: /bin/bash
])
"set_example" = toset([
"me-central-1a",
"me-central-1b",
])
"tuple_example" = [
"web-1",
4,
true,
]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
"mutated_tuple" = toset([
"4",
"true",
"web-1",
"web-2",
])
"original_tuple" = [
"web-1",
4,
true,
]
}
optional_tag = {
"Custom" = "dev"
"Name" = "web-server"
}
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
"backup_enabled" = false
"instance_type" = "t3.micro"
"monitoring" = true
"name" = "web-server"
"storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
"Environment" = "dev"
"Owner" = "platform-team"
"Project" = "sample-app"
})
value_received = "hello"
→ /workspaces/Lab11 (main) $
Lab 11
XAMPP
1:07 AM
12/31/2025

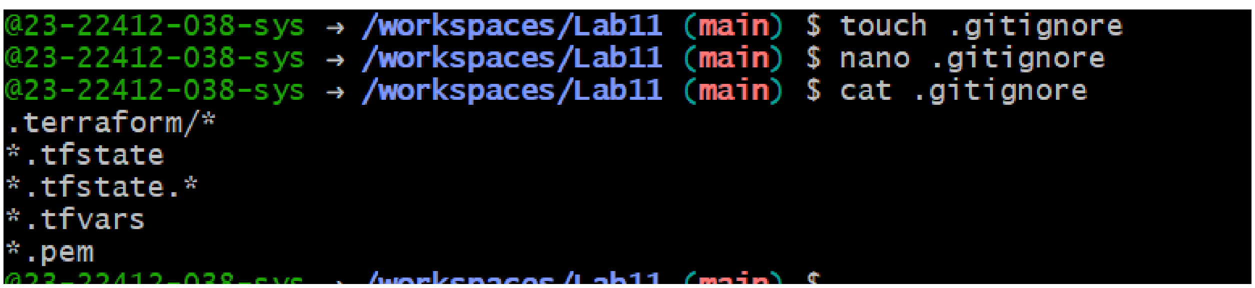

codespace@codespaces-79c51f: /bin/bash
])
"set_example" = toset([
"me-central-1a",
"me-central-1b",
])
"tuple_example" = [
"web-1",
4,
true,
]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
"mutated_tuple" = toset([
"4",
"true",
"web-1",
"web-2",
])
"original_tuple" = [
"web-1",
4,
true,
]
}
optional_tag = {
"Custom" = "dev"
"Name" = "web-server"
}
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
"backup_enabled" = false
"instance_type" = "t3.micro"
"monitoring" = true
"name" = "web-server"
"storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
"Environment" = "dev"
"Owner" = "platform-team"
"Project" = "sample-app"
})
value_received = "42"
→ /workspaces/Lab11 (main) $
Lab 11
XAMPP
1:09 AM
12/31/2025

```
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
})
value_received = [
  "a",
  "b",
  "c",
]
@23-22412-038-sys → /workspaces/Lab11 (main) $
```

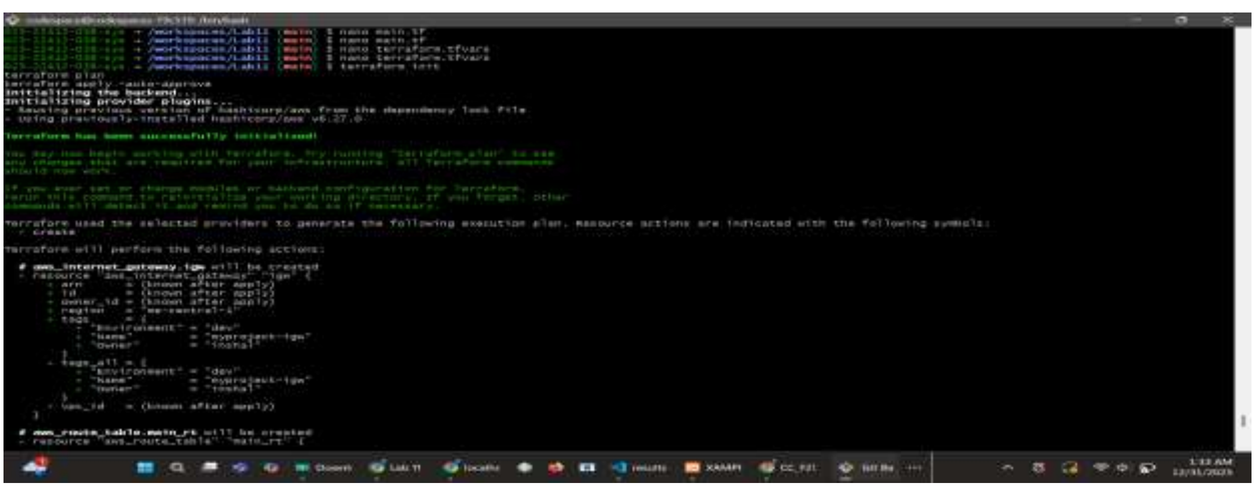
```
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}
primary_public_subnet = "subnet-08a73f8aa1dee72d3"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
})
value_received = {
  "cpu" = 4
  "name" = "server"
}
@23-22412-038-sys → /workspaces/Lab11 (main) $
```

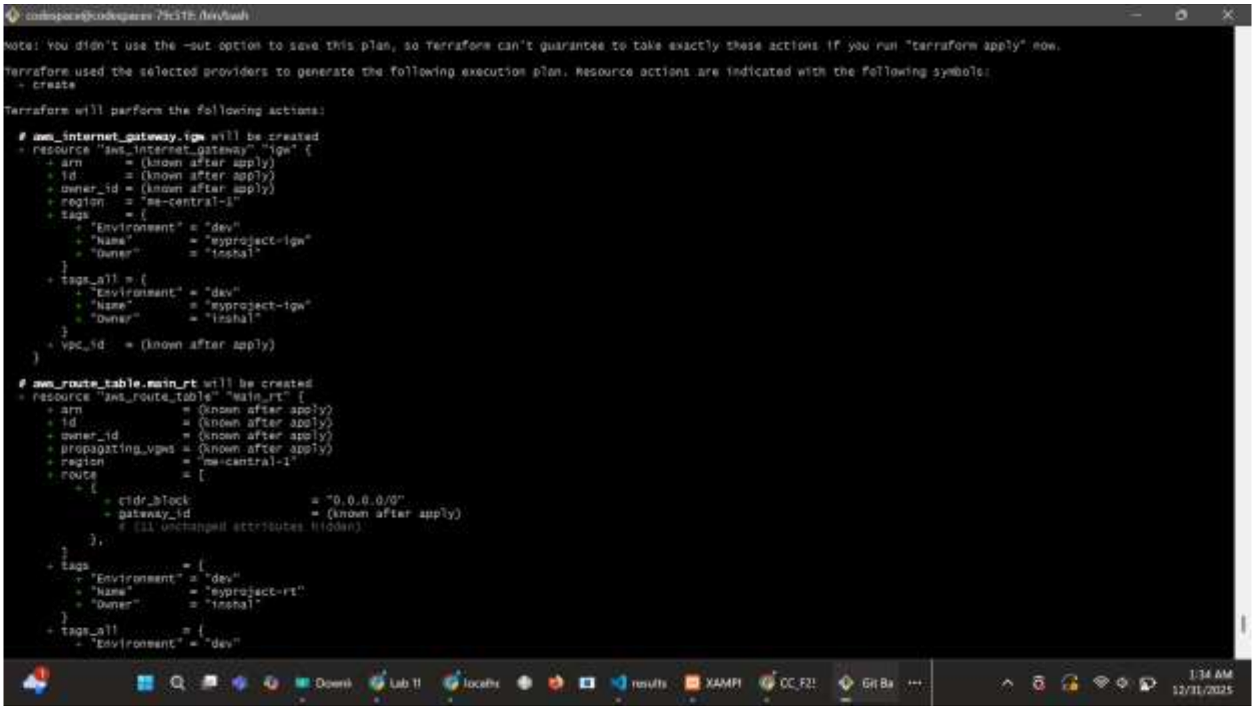


Task 7

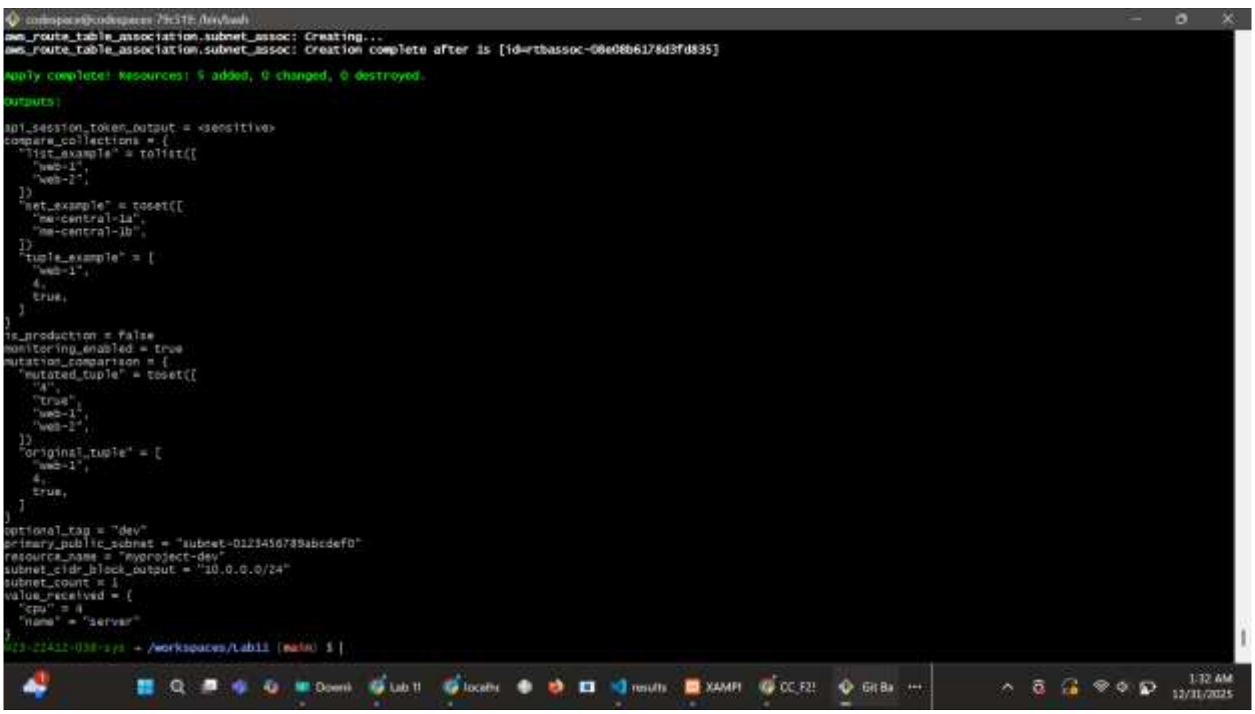

@23-22412-038-sys → /workspaces/Lab11 (main) $ touch .gitignore
@23-22412-038-sys → /workspaces/Lab11 (main) $ nano .gitignore
@23-22412-038-sys → /workspaces/Lab11 (main) $ cat .gitignore
.terraform/*
*.tfstate
.tfstate.
*.tfvars
*.pem

Task 8





```
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_internet_gateway.igw will be created
  + resource "aws_internet_gateway" "igw" {
      + arn      = (known after apply)
      + id       = (known after apply)
      + owner_id = (known after apply)
      + region   = "me-central-1"
      + tags     = {
          + "Environment" = "dev"
          + "Name"        = "myproject-igw"
          + "Owner"       = "inshal"
        }
      + tags_all = {
          + "Environment" = "dev"
          + "Name"        = "myproject-igw"
          + "Owner"       = "inshal"
        }
      + vpc_id   = (known after apply)
    }

  # aws_route_table.main_rt will be created
  + resource "aws_route_table" "main_rt" {
      + arn              = (known after apply)
      + id               = (known after apply)
      + owner_id         = (known after apply)
      + propagating_vgws = (known after apply)
      + region           = "me-central-1"
      + route            = [
          + {
              + cidr_block                 = "0.0.0.0/0"
              + gateway_id                 = (known after apply)
                # (11 unchanged attributes hidden)
            },
        ]
      + tags             = {
          + "Environment" = "dev"
          + "Name"        = "myproject-rt"
          + "Owner"       = "inshal"
        }
      + tags_all         = {
          + "Environment" = "dev"
```



```
aws_route_table_association.subnet_assoc: Creating...
aws_route_table_association.subnet_assoc: Creation complete after 1s [id=rtbassoc-08e08b6178d3fd835]

Apply complete! Resources: 5 added, 0 changed, 0 destroyed.

Outputs:

api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = "dev"
primary_public_subnet = "subnet-0123456789abcdef0"
resource_name = "myproject-dev"
subnet_cidr_block_output = "10.0.0.0/24"
subnet_count = 1
value_received = {
  "cpu" = 4
  "name" = "server"
}
```

Task 9

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ terraform init
terraform apply -auto-approve
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
aws_route_table_association.subnet_assoc: Refreshing state... [id=rtbassoc-08e08b6178d3fd835]
aws_vpc.main_vpc: Refreshing state... [id=vpc-0cd12f95de8556417]
aws_subnet.app_subnet: Refreshing state... [id=subnet-0b1696fda58ee7a71]
aws_internet_gateway.igw: Refreshing state... [id=igw-078d80b905e5f9c60]
aws_route_table.main_rt: Refreshing state... [id=rtb-079ccd3c3c57d9571]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create
  - destroy

Terraform will perform the following actions:

  # aws_default_route_table.main_rt will be created
  + resource "aws_default_route_table" "main_rt" {
      + arn                    = (known after apply)
      + default_route_table_id = (known after apply)
      + id                     = (known after apply)
      + owner_id               = (known after apply)
      + region                 = "me-central-1"
```

```
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create
  - destroy

Terraform will perform the following actions:

  # aws_default_route_table.main_rt will be created
  + resource "aws_default_route_table" "main_rt" {
      + arn                    = (known after apply)
      + default_route_table_id = (known after apply)
      + id                     = (known after apply)
      + owner_id               = (known after apply)
      + region                 = "me-central-1"
      + route                  = [
          + {
              + cidr_block                 = "0.0.0.0/0"
              + gateway_id                 = (known after apply)
                # (10 unchanged attributes hidden)
            },
        ]
      + tags                   = {
          + "Name" = "dev-rt"
        }
      + tags_all               = {
          + "Name" = "dev-rt"
        }
      + vpc_id                 = (known after apply)
    }

  # aws_default_security_group.myapp_sg will be created
  + resource "aws_default_security_group" "myapp_sg" {
      + arn                    = (known after apply)
      + description            = (known after apply)
      + egress                 = [
          + {
              + cidr_blocks      = [
                  + "0.0.0.0/0",
                ]
              + from_port        = 0
              + ipv6_cidr_blocks = []
              + prefix_list_ids  = []
              + protocol         = "-1"
              + security_groups  = []
              + self             = false
              + to_port          = 0
                # (1 unchanged attribute hidden)
```

```
          + self              = false
          + to_port           = 0
            # (1 unchanged attribute hidden)
        },
    ]
  + id                        = (known after apply)
  + ingress                   = [
      + {
          + cidr_blocks       = [
              + "0.0.0.0/0",
            ]
          + from_port         = 80
          + ipv6_cidr_blocks  = []
          + prefix_list_ids   = []
          + protocol          = "tcp"
          + security_groups   = []
          + self              = false
          + to_port           = 80
            # (1 unchanged attribute hidden)
        },
      + {
          + cidr_blocks       = [
              + "4.240.18.229/32",
            ]
          + from_port         = 22
          + ipv6_cidr_blocks  = []
          + prefix_list_ids   = []
          + protocol          = "tcp"
          + security_groups   = []
          + self              = false
          + to_port           = 22
            # (1 unchanged attribute hidden)
        },
    ]
  + name                      = (known after apply)
  + name_prefix               = (known after apply)
  + owner_id                  = (known after apply)
  + region                    = "me-central-1"
  + revoke_rules_on_delete    = false
  + tags                      = {
      + "Name" = "dev-sg"
    }
  + tags_all                  = {
      + "Name" = "dev-sg"
    }
  + vpc_id                    = (known after apply)
}
```

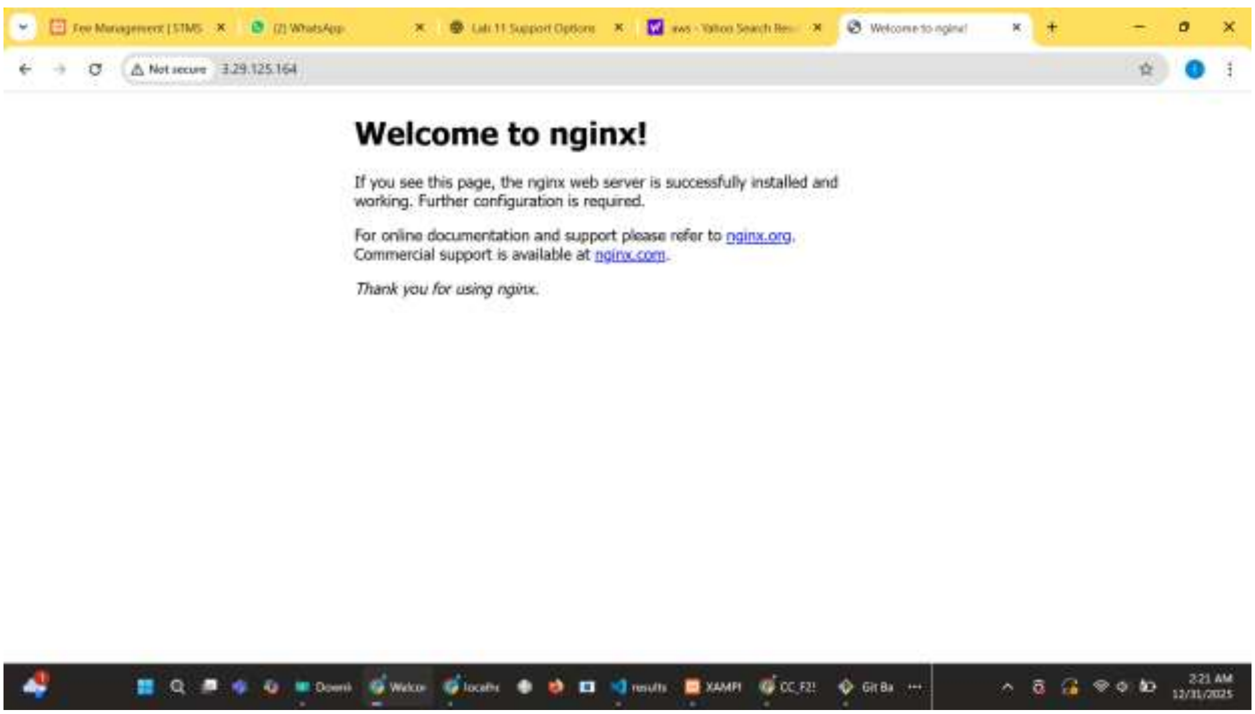
```
# aws_internet_gateway.myapp_igw will be created
+ resource "aws_internet_gateway" "myapp_igw" {
    + arn      = (known after apply)
    + id       = (known after apply)
    + owner_id = (known after apply)
    + region   = "me-central-1"
    + tags     = {
        + "Name" = "dev-igw"
      }
    + tags_all = {
        + "Name" = "dev-igw"
      }
    + vpc_id   = (known after apply)
  }

# aws_key_pair.ssh_key will be created
+ resource "aws_key_pair" "ssh_key" {
    + arn             = (known after apply)
    + fingerprint     = (known after apply)
    + id              = (known after apply)
    + key_name        = "serverkey"
    + key_name_prefix = (known after apply)
    + key_pair_id     = (known after apply)
    + key_type        = (known after apply)
    + public_key      = "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPNwu9FBKoFK2EhsRwTKXLcwtL0kr5yY8EtDI/qQ+THD codespace@codespaces-79c519"
    + region          = "me-central-1"
    + tags_all        = (known after apply)
  }
```

```
aws_route_table_association.subnet_assoc: Destroying... [id=rtbassoc-08e08b6178d3fd835]
aws_key_pair.ssh_key: Creating...
aws_vpc.myapp_vpc: Creating...
aws_key_pair.ssh_key: Creation complete after 0s [id=serverkey]
aws_route_table_association.subnet_assoc: Destruction complete after 1s
aws_route_table.main_rt: Destroying... [id=rtb-079ccd3c3c57d9571]
aws_subnet.app_subnet: Destroying... [id=subnet-0b1696fda58ee7a71]
aws_subnet.app_subnet: Destruction complete after 0s
aws_route_table.main_rt: Destruction complete after 0s
aws_internet_gateway.igw: Destroying... [id=igw-078d80b905e5f9c60]
aws_internet_gateway.igw: Destruction complete after 1s
aws_vpc.main_vpc: Destroying... [id=vpc-0cd12f95de8556417]
aws_vpc.myapp_vpc: Creation complete after 2s [id=vpc-08c0de79b0e26507d]
aws_subnet.myapp_subnet_1: Creating...
aws_internet_gateway.myapp_igw: Creating...
aws_default_security_group.myapp_sg: Creating...
aws_vpc.main_vpc: Destruction complete after 0s
aws_internet_gateway.myapp_igw: Creation complete after 1s [id=igw-08deab4a9bafbaac7]
aws_default_route_table.main_rt: Creating...
aws_subnet.myapp_subnet_1: Creation complete after 1s [id=subnet-03e5bda232f8fa581]
aws_default_route_table.main_rt: Creation complete after 0s [id=rtb-0e0e0308f83c69ff2]
aws_default_security_group.myapp_sg: Creation complete after 3s [id=sg-0a106f65daa2ea6eb]
aws_instance.myapp_server: Creating...
aws_instance.myapp_server: Still creating... [00m10s elapsed]
aws_instance.myapp_server: Creation complete after 12s [id=i-097e723716367b59c]
```

```
Apply complete! Resources: 7 added, 0 changed, 5 destroyed.

Outputs:

ec2_public_ip = "3.29.125.164"
subnet_id = "subnet-03e5bda232f8fa581"
vpc_id = "vpc-08c0de79b0e26507d"
```

```
@23-22412-038-sys → /workspaces/Lab11 (main) $ ssh -i ~/.ssh/id_ed25519 ec2-user@3.29.125.164
The authenticity of host '3.29.125.164 (3.29.125.164)' can't be established.
ED25519 key fingerprint is SHA256:DSNFt1nKTtnAA2a+sFA9RRoOue+1eUoM5LssKxBVIXo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.29.125.164' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-10-168 ~]$
```


[ec2-user@ip-10-0-10-168 ~]$ curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>
<p>For online documentation and support please refer to
nginx.org.
Commercial support is available at
nginx.com.</p>
<p>Thank you for using nginx.</p>
</body>
</html>


Not secure 3.29.125.164
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.

Cleanup

```
023-22412-038-sys → /workspaces/Lab11 (main) $ terraform destroy -auto-approve
aws_key_pair.ssh_key: Refreshing state... [id=serverkey]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-08c0de79b0e26507d]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-08deab4a9bafbaac7]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-03e5bda232f8fa581]
aws_default_security_group.myapp_sg: Refreshing state... [id=sg-0a106f65daa2ea6eb]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-0e0e0308f83c69ff2]
aws_instance.myapp_server: Refreshing state... [id=i-097e723716367b59c]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # aws_default_route_table.main_rt will be destroyed
  - resource "aws_default_route_table" "main_rt" {
      - arn                    = "arn:aws:ec2:me-central-1:623705167686:route-table/rtb-0e0e0308f83c69ff2" -> null
      - default_route_table_id = "rtb-0e0e0308f83c69ff2" -> null
      - id                     = "rtb-0e0e0308f83c69ff2" -> null
      - owner_id               = "623705167686" -> null
      - propagating_vgws       = [] -> null
      - region                 = "me-central-1" -> null
      - route                  = [
          - {
              - cidr_block                 = "0.0.0.0/0"
              - gateway_id                 = "igw-08deab4a9bafbaac7"
                # (10 unchanged attributes hidden)
            },
        ] -> null
      - tags                   = {
          - "Name" = "dev-rt"
        } -> null
      - tags_all               = {
          - "Name" = "dev-rt"
        } -> null
      - vpc_id                 = "vpc-08c0de79b0e26507d" -> null
    }

  # aws_default_security_group.myapp_sg will be destroyed
  - resource "aws_default_security_group" "myapp_sg" {
      - arn                    = "arn:aws:ec2:me-central-1:623705167686:security-group/sg-0a106f65daa2ea6eb" -> null
      - description            = "default VPC security group" -> null
```

```
# aws_default_security_group.myapp_sg will be destroyed
- resource "aws_default_security_group" "myapp_sg" {
    - arn                    = "arn:aws:ec2:me-central-1:623705167686:security-group/sg-0a106f65daa2ea6eb" -> null
    - description            = "default VPC security group" -> null
    - egress                 = [
        - {
            - cidr_blocks      = [
                - "0.0.0.0/0",
              ]
            - from_port        = 0
            - ipv6_cidr_blocks = []
            - prefix_list_ids  = []
            - protocol         = "-1"
            - security_groups  = []
            - self             = false
            - to_port          = 0
```

```
# aws_instance.myapp_server will be destroyed
- resource "aws_instance" "myapp_server" {
    - ami                                  = "ami-05524d6658fcf35b6" -> null
    - arn                                  = "arn:aws:ec2:me-central-1:623705167686:instance/i-097e723716367b59c" -> null
    - associate_public_ip_address          = true -> null
    - availability_zone                    = "me-central-1a" -> null
    - disable_api_stop                     = false -> null
    - disable_api_termination              = false -> null
    - ebs_optimized                        = false -> null
    - force_destroy                        = false -> null
    - get_password_data                    = false -> null
    - hibernation                          = false -> null
    - id                                   = "i-097e723716367b59c" -> null
    - instance_initiated_shutdown_behavior = "stop" -> null
    - instance_state                       = "running" -> null
    - instance_type                        = "t3.micro" -> null
    - ipv6_address_count                   = 0 -> null
    - ipv6_addresses                       = [] -> null
    - key_name                             = "serverkey" -> null
    - monitoring                           = false -> null
    - placement_partition_number           = 0 -> null
    - primary_network_interface_id         = "eni-05c432978c007a7ce" -> null
    - private_dns                          = "ip-10-0-10-168.me-central-1.compute.internal" -> null
    - private_ip                           = "10.0.10.168" -> null
    - public_ip                            = "3.29.125.164" -> null
    - region                               = "me-central-1" -> null
    - secondary_private_ips                = [] -> null
    - security_groups                      = [] -> null
    - source_dest_check                    = true -> null
    - subnet_id                            = "subnet-03e5bda232f8fa581" -> null
    - tags                                 = {
        - "Name" = "dev-ec2-instance"
      } -> null
    - tags_all                             = {
        - "Name" = "dev-ec2-instance"
      } -> null
    - tenancy                              = "default" -> null
    - user_data                            = <<-EOT
          #!/bin/bash
          yum update -y
          yum install -y nginx
          systemctl start nginx
```

```
# aws_key_pair.ssh_key will be destroyed
- resource "aws_key_pair" "ssh_key" {
    - arn             = "arn:aws:ec2:me-central-1:623705167686:key-pair/serverkey" -> null
    - fingerprint     = "ECNX8m9VY35PKyVYXh9KaRj4J4wAqSwqyQfVd/1ZtXo=" -> null
    - id              = "serverkey" -> null
    - key_name        = "serverkey" -> null
    - key_pair_id     = "key-045da6b074c103868" -> null
    - key_type        = "ed25519" -> null
    - public_key      = "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPNwu9F8KoFKzEhsRwTKKLcwtL0kr5yY8EtDI/qQ+THD codespace@codespaces-79c519" -> null
    - region          = "me-central-1" -> null
    - tags            = {} -> null
    - tags_all        = {} -> null
```

```
# aws_subnet.myapp_subnet_1 will be destroyed
- resource "aws_subnet" "myapp_subnet_1" {
    - arn                                            = "arn:aws:ec2:me-central-1:623705167686:subnet/subnet-03e5bda232f8fa581" -> null
    - assign_ipv6_address_on_creation                = false -> null
    - availability_zone                              = "me-central-1a" -> null
    - availability_zone_id                           = "mec1-az1" -> null
    - cidr_block                                     = "10.0.10.0/24" -> null
    - enable_dns64                                   = false -> null
    - enable_lni_at_device_index                     = 0 -> null
    - enable_resource_name_dns_a_record_on_launch    = false -> null
    - enable_resource_name_dns_aaaa_record_on_launch = false -> null
    - id                                             = "subnet-03e5bda232f8fa581" -> null
    - ipv6_native                                    = false -> null
    - map_customer_owned_ip_on_launch                = false -> null
    - map_public_ip_on_launch                        = false -> null
    - owner_id                                       = "623705167686" -> null
    - private_dns_hostname_type_on_launch            = "ip-name" -> null
    - region                                         = "me-central-1" -> null
```

```
Plan: 0 to add, 0 to change, 7 to destroy.

Changes to Outputs:
  - ec2_public_ip = "3.29.125.164" -> null
  - subnet_id     = "subnet-03e5bda232f8fa581" -> null
  - vpc_id        = "vpc-08c0de79b0e26507d" -> null
aws_default_route_table.main_rt: Destroying... [id=rtb-0e0e0308f83c69ff2]
aws_instance.myapp_server: Destroying... [id=i-097e723716367b59c]
aws_default_route_table.main_rt: Destruction complete after 0s
aws_internet_gateway.myapp_igw: Destroying... [id=igw-08deab4a9bafbaac7]
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-08deab4a9bafbaac7, 00m10s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m20s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-08deab4a9bafbaac7, 00m20s elapsed]
aws_internet_gateway.myapp_igw: Destruction complete after 27s
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m30s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-097e723716367b59c, 00m40s elapsed]
aws_instance.myapp_server: Destruction complete after 40s
aws_key_pair.ssh_key: Destroying... [id=serverkey]
aws_subnet.myapp_subnet_1: Destroying... [id=subnet-03e5bda232f8fa581]
aws_default_security_group.myapp_sg: Destroying... [id=sg-0a106f65daa2ea6eb]
aws_default_security_group.myapp_sg: Destruction complete after 0s
aws_key_pair.ssh_key: Destruction complete after 0s
aws_subnet.myapp_subnet_1: Destruction complete after 0s
aws_vpc.myapp_vpc: Destroying... [id=vpc-08c0de79b0e26507d]
aws_vpc.myapp_vpc: Destruction complete after 1s
```

```
Destroy complete! Resources: 7 destroyed.
@23-22412-038-sys → /workspaces/Lab11 (main) $ |
```